

CHE 201 Fall 2019 Quiz 3

Paul Diaz

TOTAL POINTS

95 / 100

QUESTION 1

11 10 / 15

- **0 pts** Correct
- **5 pts** first one is wrong
- ✓ - **5 pts** second one is wrong
- **5 pts** third one is wrong

QUESTION 2

22 85 / 85

- ✓ - **0 pts** Correct
- **15 pts** mass balance is incorrect
- **20 pts** energy balance is incorrect
- **10 pts** h1 is incorrect
- **10 pts** h2 is incorrect
- **10 pts** h3 is incorrect
- **15 pts** m3 is wrong

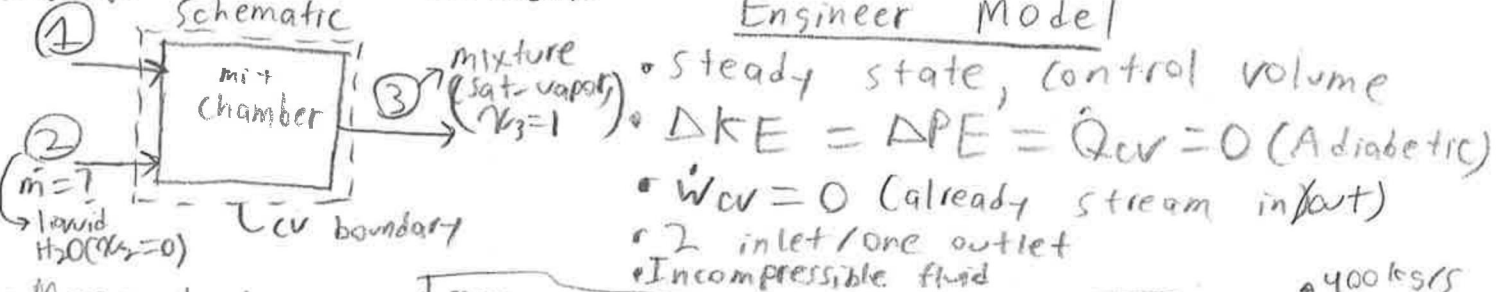
NAME: Paul DiazUIN: 662973726**QUIZ 3: (100 pts) (Close book/closed notes) Show all your work for full credit**1. (15 points) State whether the following statements are **True** or **False** (5 pts each)True A thermodynamic cycle cannot convert heat completely into work. *Kelvin-Planck ✓*True The change in entropy (ΔS) for a system cannot be less than zero. *yes for system + surroundings must increase*False A closed system process that violates the 2nd law of thermodynamics must also violate the 1st law of thermodynamics.*conservation of NRG → entropy of universe always increases but 1st law is always true*

2. (85 points) A stream of superheated steam at 3 bar and 280 °C with a mass flow rate of 400 kg/s (stream 1) is mixed with a stream of liquid water at 3 bar and 20 °C (stream 2) in an adiabatic mixing chamber. The resulting mixture exits as a saturated vapor at 3 bar (stream 3). Assume steady state operation and neglect any changes in kinetic and potential energy. Calculate the mass flow rate of the liquid water (stream 2) entering the mixing chamber, in kg/s.

Solve

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_i \dot{m}_i \left(h_i + \frac{V_i^2}{2} + g z_i \right) - \sum_e \dot{m}_e \left(h_e + \frac{V_e^2}{2} + g z_e \right) \quad \dot{m} = \frac{AV}{v}$$

1 joule = 1 N.m
1 N = 1 kg.m/s²

Engineer Model

Mass balance: $\frac{dm_{cv}}{dt} = 0 = \dot{m}_1 + \dot{m}_2 - \dot{m}_3 \Rightarrow \dot{m}_3 = \dot{m}_1 + \dot{m}_2$

Energy balance: $\frac{dE}{dt} = 0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1 \left(h_1 + \frac{V_1^2}{2} + g z_1 \right) + \dot{m}_2 (h_2) - \dot{m}_3 (h_3)$

$$0 = \dot{m}_1 (h_1) + \dot{m}_2 (h_2) - \dot{m}_3 (h_3)$$

$$h_1 = h(3 \text{ bar}, 280^\circ\text{C}) = 3028.6 \frac{\text{kJ}}{\text{kg}} \quad h_2 = h_f(3 \text{ bar}, 20^\circ\text{C}) \approx h_f(20^\circ\text{C})$$

$$h_3 = h_g(3 \text{ bar}, T_{\text{sat}}) = 2725.3 \frac{\text{kJ}}{\text{kg}} \quad h_2 \approx h_f(20^\circ\text{C}) = 83.96 \frac{\text{kJ}}{\text{kg}}$$

$$0 = \dot{m}_1 (h_1) + \dot{m}_2 (h_2) - (\dot{m}_1 + \dot{m}_2) (h_3) = \dot{m}_1 (h_1 - h_3) + \dot{m}_2 (h_2 - h_3)$$

$$-\dot{m}_2 (h_2 - h_3) = \dot{m}_1 (h_1 - h_3) \Rightarrow \dot{m}_2 = -\dot{m}_1 \frac{(h_1 - h_3)}{(h_2 - h_3)} = -400 \frac{\text{kg/s}}{83.96 \frac{\text{kJ}}{\text{kg}} - 2725.3 \frac{\text{kJ}}{\text{kg}}} = 45.9 \frac{\text{kg/s}}$$

